

Pranav Senthilkumar

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EDUCATION

Texas A&M University

Bachelor of Engineering, Engineering Honors

College Station, TX

Aug. 2025 – May 2029

EXPERIENCE

Student Researcher

Texas A&M University

December 2025-Present

College Station, TX

- Architecting a sonar-based sensing system for subsurface evaluation of bridge foundation stability (TxDOT collaboration), focusing on signal acquisition, processing, and geospatial integration
- Developing a scalable pipeline for processing echo sounder and side-scan sonar data, including signal conditioning, feature extraction, and visualization of underwater terrain
- Exploring machine learning methods for anomaly detection in sonar signals to identify subsurface movement and geotechnical instability

Researcher Team Lead

AlgoVerse AI Research

June 2024 – October 2024

Remote

- Conducted applied research on calibration and alignment of large language models in ambiguous real-world decision contexts
- Fine-tuned transformer models (Llama-3, Mistral-7B, Zephyr-7B) using QLoRA, achieving up to 31% improvement in Dirichlet loss
- Designed evaluation pipelines using ethical reasoning datasets to benchmark probabilistic alignment and model behavior under uncertainty

SKILLS

Programming Languages: Python, C++, Java

Scientific / Data Libraries: NumPy, Pandas, OpenCV, MediaPipe

AI / ML: TensorFlow, PyTorch, Keras, SciKit Learn

Systems & Tools: GitHub, Firebase, Arduino, KiCad (learning), Microsoft Flight Simulator

Productivity Tools: Microsoft Office, LaTeX

PROJECTS

Haptic Portal

Python, DepthAI, OpenCV, MediaPipe, UDP, KICAD

January 2026 – Present

- Developing a real-time vision-to-haptics system that converts hand tracking and Oak-D-Lite camera stereo depth sensing into tactile control signals for interactive haptic devices
- Implemented a depth-processing module that converts stereo depth maps into a compact 5×5 feature matrix, reducing high-resolution depth data into 25 normalized channels for real-time control
- Added temporal filtering and median smoothing to stabilize hand-depth measurements across frames, improving reliability of the real-time haptic signal
- Designing a PCB integrating a Raspberry Pi Pico and PCA9685 drivers, with a structured layout and managed power routing for reliable operation

AI-Based Push-Up Form Feedback System

Python, OpenCV, MediaPipe, NumPy

June 2025 – July 2025

- Engineered a custom push-up detection algorithm using MediaPipe pose landmarks, calculating joint angles and filtering noise for accurate rep counting and posture analysis
- Built a back angle classification system with vector math between scapular, hip, and knee keypoints to assess form accuracy and give corrective feedback
- Implemented orientation correction logic to adapt pose estimation across varied camera angles, improving model generalization
- Designed a real-time visual feedback overlay using OpenCV to display posture ratings and guide proper technique

PUBLICATIONS

- Pranav Senthilkumar, et al. *Fine-Tuning Language Models for Ethical Ambiguity: A Comparative Study of Alignment with Human Responses*. Accepted at NeurIPS 2024, SoLaR Workshop. arXiv:2410.07826